

Opinion

Sonographer interaction with artificial intelligence: collaboration or conflict?

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Artificial intelligence in fetal ultrasound

Recent years have seen rapid development in the use of artificial intelligence (AI) in many aspects of healthcare, especially in the field of medical imaging. In diverse specialties, including ophthalmology, radiology and cardiology, machine learning (a subtype of AI) is now either in clinical use or on the cusp of being so¹. Underlying this explosion of interest is the significant improvement in model performance, driven largely by three factors. Firstly, advances in the necessary hardware, specifically graphical processing units used to train deep machine-learning algorithms with large amounts of high-dimensional data, such as images; secondly, the development of effective AI systems and machine-learning model architectures, such as convolutional neural networks²; and thirdly, the increasing availability of large digital datasets that can be used for training. This improvement in performance has led to the approval by the United States Food and Drug Administration of over 500 AI-enabled medical devices for specific clinical tasks³. There is now a large body of research equating the performance of AI models with that of the human expert in medical imaging-based diagnostic tasks, albeit with the important caveat that certain methodological deficiencies were present in some of the studies⁴.

Fetal ultrasound has also been an area of intense AI research, with applications that could improve workflow, imaging quality or disease detection^{5–10}. Indeed, AI-based functionality is now integrated into the latest commercially available models of obstetric ultrasound machine from several vendors^{11,12}. The potential uses of AI in fetal ultrasound can be divided into three broad areas,

detailed below. Key papers and performance metrics are summarized in Table 1.

Automated fetal biometry

This is the most widely studied application of AI in fetal ultrasound imaging, and probably the only task for which AI is in current clinical use outside of a research setting. Models have been developed to automatically measure head circumference, biparietal diameter, femur length, crown–rump length and nuchal translucency thickness^{8,13–16}. These models have the potential to reduce interobserver variability, and early clinical translation studies have shown a significant time-saving effect compared with manual measurement⁸.

Automatic detection and saving of standard image planes

Models are available that can analyze each frame of a stream of ultrasound video and recognize when a specific standard sonographic plane is being imaged^{8,17,18}. As with automated biometry, significant time savings have been demonstrated, and there is also the potential to audit scan completeness by automatically labeling each frame of the scan with the fetal anatomy displayed⁸. There is also the possibility of AI giving feedback to the human sonographer about the quality of the acquired images^{9,19}.

Automatic detection of fetal disease

This is perhaps the most challenging role for AI in fetal ultrasound. There are currently no commercially available products that assist directly in the detection of fetal disease, but several papers utilizing retrospective data have been published with encouraging results^{10,20–22}.

Despite these exciting advances in research, it remains unclear exactly how AI will be integrated into sonographer clinical workflow in the future, and there are a number of challenges specific to obstetric ultrasound to overcome. Unlike most imaging modalities, ultrasound examination entails simultaneous image acquisition and interpretation, meaning that any AI system will likely have to operate in real time, providing appropriate feedback to the sonographer as they perform the scan. In this Opinion, we explore some of the most pressing issues that require further investigation and outline potential pitfalls and challenges in this field.

Are three eyes better than two? Human–AI collaboration

Unlike many other medical-imaging tasks, the real-time nature of ultrasound, with simultaneous image acquisition

and analysis, means that complete automation of these processes is unlikely to be achieved in the short or medium term. To further complicate matters, there is a wide variation in the number and sequence of fetal structures imaged during a scan, with sonographers taking advantage of fetal position and other factors to optimize scan images²³. This lack of consistency and huge variation in image appearance, even between images of the same subject taken by different operators, makes the application of AI more challenging. For these reasons, the use of AI in fetal ultrasound will almost certainly fall on the spectrum between full human agency and full automation, as is the case for many other complex non-medical tasks (Figure 1)²⁴.

During an ultrasound scan, both the human ultrasound operator and the AI system are receiving inputs and producing outputs simultaneously, so some kind of interaction between the two will be necessary. This is sometimes referred to as the ‘human-in-the-loop’ approach. Both humans and AI will be operating under a certain level of uncertainty and/or be susceptible to biases, and both may demonstrate unpredictable behaviors that could be undesirable or even dangerous²⁵. We can conceptualize the human–AI team as possessing a ‘hybrid intelligence’ (Figure 2), and the goal must be for this hybrid intelligence to have superior performance within a given task compared with the human or AI alone²⁶. This

Table 1 Selection of key papers evaluating use of artificial intelligence (AI) in fetal ultrasound scanning and corresponding performance metrics

Application category	Summary
Automated biometry	
Sinclair <i>et al.</i> ¹³	A fully convolutional neural network was trained to perform semantic segmentation of the fetal head, using a dataset of around 2000 manually segmented neurosonographic images obtained during routine clinical scans. The model matched human expert-level performance based on model–expert and expert–expert Dice coefficient (0.98), and could be run in real time at 15 frames/s.
Salim <i>et al.</i> ¹⁴	A commercially developed model that automatically measures head circumference, abdominal circumference and femur length on two-dimensional fetal ultrasound images was assessed in 100 fetuses, and compared with expert manual measurement. In 53.3% of images, automated calipers were judged to be in an acceptable position, but a major or minor adjustment was required in 35.5% and 10.1% of images, respectively. The mean percentage error between model and expert was between −0.81% and 3.76%, depending on the biometric parameter, which is comparable with human–human interobserver variability.
Automated plane detection	
Baumgartner <i>et al.</i> ¹⁷	A convolutional neural network was trained to classify two-dimensional fetal ultrasound images as either one of 12 standard planes or background, using a dataset of 1003 ultrasound scans. Working in real time, the model detected standard planes with a precision of 69% and recall of 80%.
Cai <i>et al.</i> ¹⁸	A multitask convolutional neural network was trained using 1616 images from 33 fetal ultrasound scans, with each frame labeled as abdominal circumference plane or background. The model was also trained, using sonographer eye-tracking data, to predict visual attention for a given ultrasound image. Combining the ultrasound image with an estimated visual attention map improved classification performance, giving a precision of 99% and recall of 96%.
Automated diagnosis	
Arnaout <i>et al.</i> ¹⁰	An ensemble of neural networks was trained, using a dataset of 107 823 images from 1326 echocardiograms, to classify fetal ultrasound images into standard cardiac planes and discriminate between normal hearts and those with congenital heart disease (CHD). In an internal test set, the ensemble achieved 95% sensitivity, 96% specificity and area under the receiver-operating-characteristics curve of 0.99 in detecting CHD.
Wang <i>et al.</i> ²⁰	Various models were trained to automatically segment and measure the post-left atrium space, using a dataset of 642 images from 319 fetuses. This measurement was used to discriminate between normal fetuses and those with total anomalous pulmonary venous drainage, with 89.5% sensitivity and 86.6% specificity in the best performing model.

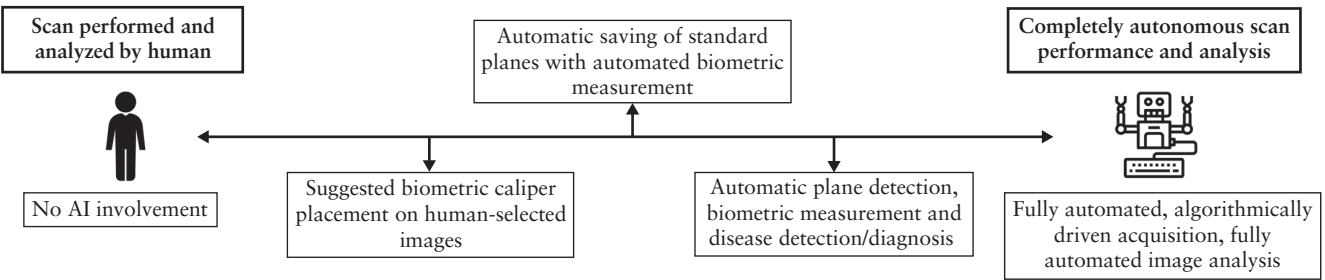


Figure 1 Spectrum of possible involvement of artificial intelligence (AI) in fetal ultrasound imaging, from no AI involvement to full scan automation.

has been demonstrated for some medical tasks, such as endoscopy interpretation, chest X-ray diagnosis and skin cancer detection, but has not been studied in detail for fetal ultrasound^{26–28}.

Several steps are required before sonographer–AI collaboration can be seen as effective: (1) the AI model needs to demonstrate the ability to meet or exceed human-level performance, according to quantitative metrics; (2) the sonographer needs to trust and utilize the output of the AI model when it is correct; and (3) importantly, the sonographer needs to recognize when the AI output is incorrect, and overrule it. An additional stage in which the AI model recognizes human error and alerts the operator may also be desirable. Failure at any of these stages may result in a sonographer–AI team that has inferior performance to that of human sonographers alone. This is important to recognize because, if left unchecked, the introduction of AI could lead to worse care for patients.

A key aspect of human–AI collaboration is trust²⁹. Some authorities have argued that the concept of trust is inappropriate in this context, either because it is conceptually confused (we can never trust AI, because AI is not capable of ‘trustworthiness’ in the human sense) or dangerous (we should not trust AI, because we can never be sure it is acting as intended)^{30–32}. Regardless of semantics, trust, or at least the assessment by the human of the reliability of AI, is important to consider. It is

known that users are more likely to use an AI system that they trust and abandon its use when their confidence in the system’s reliability goes down³³. This course of action is appropriate when in response to a true deterioration in AI performance, but problematic when the human ignores the outputs of a high-performing AI model (so called ‘algorithm aversion’). Conversely, excessive trust in an AI system that is performing poorly can be even more dangerous in a medical context (so-called ‘automation bias’); for example, blindly trusting automatic fetal biometric measurements that are erroneous would result in an inaccurate scan report.

Breakdown of trust: algorithm aversion

The relationship between sonographers and AI has a potentially difficult start, as it is well documented that patients and doctors prefer human predictions over those from a computer^{34,35}. Interestingly, this preference persists even after it has been demonstrated that the computer-based predictions are superior^{34,36}. The causes of this phenomenon are not fully understood, but may relate to a belief that humans can better understand the reasons underlying a decision made by a fellow human, and so are more confident in assessing the likely accuracy of that prediction³⁷. It is also important for human operators to feel that the technology is assisting them

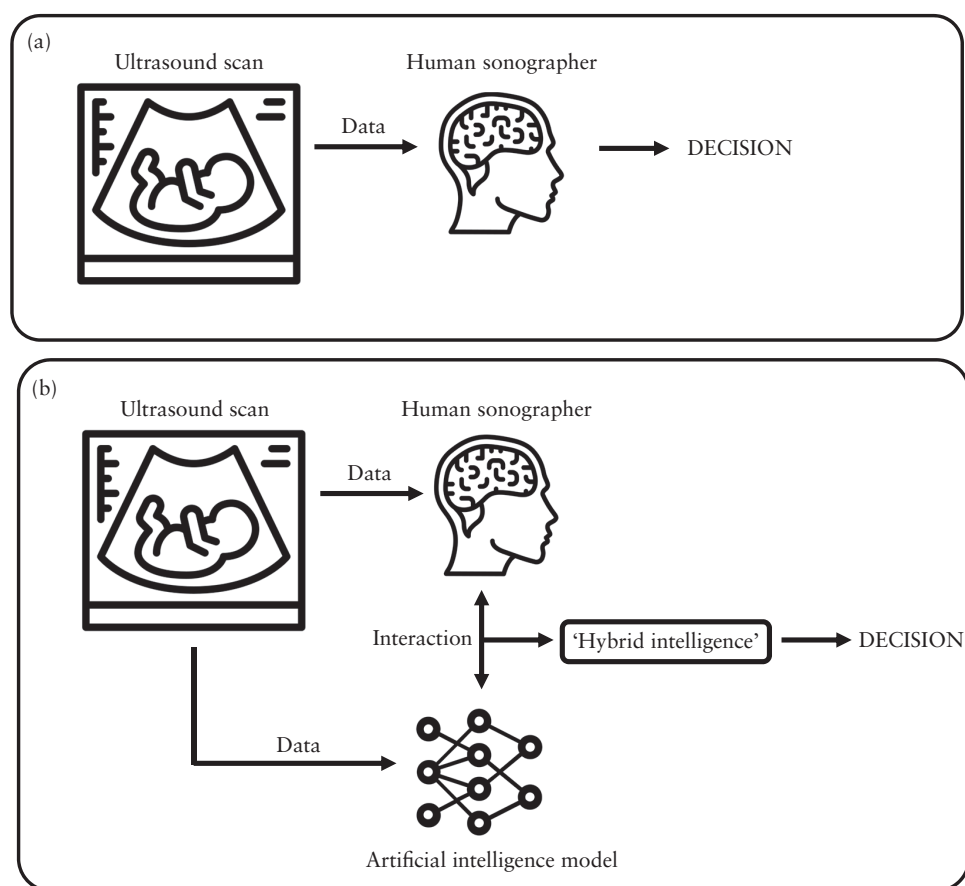


Figure 2 Data flow and decision-making in: (a) traditional manual ultrasound scanning and (b) ultrasound scanning with assistance from artificial intelligence.

or making their work easier; studies have shown that novel technology is likely to be abandoned quickly if it is felt that it does not add value, or if the functionality can be obtained in other ways³⁸.

In addition to the baseline disadvantage that AI faces in gaining trust, humans judge AI harshly when they observe it to have made a single error and are more likely to abandon its use compared with witnessing occasional human errors³⁶. It is likely that all clinicians have observed a colleague making some kind of error in interpretation or diagnosis, but it would not be rational to disregard all decisions made by that colleague from that point onwards. However, this behavior is well-described after an AI error³⁶. This has important implications for the integration of AI into a clinical environment.

An issue that is often raised when discussing loss of trust in AI systems is that of 'explainability'³⁹. The complexity of modern deep-learning models means that it has become extremely difficult to define exactly why a model has made a prediction based on given data. This lack of explainability has been termed the 'black-box problem' and may pose a barrier to the clinical translation of AI^{40,41}. Clinicians are trained to make decisions based on available data, and if the only data available to them is the output of an AI model, with no means of assessing whether this output is correct (or even sensible), it becomes difficult for them to trust the AI model. It is also difficult to detect biases in AI outputs, meaning that the human operator may unknowingly propagate these biases, exposing them to criticism. Improving explainability is not a trivial task, and there is little evidence that it improves the willingness of humans to use an AI system⁴¹. Despite this, it is likely that successful AI-based medical products in the future will have at least some degree of explainability, not least because this requirement is becoming codified into international legal frameworks⁴².

Too much trust: automation bias

A breakdown in trust between the sonographer and the AI system, resulting in the AI output being disregarded, would likely revert the overall performance standard to that of an independently operating human. A potentially more dangerous situation in this context is an excess of trust, leading the sonographer to act on erroneous AI outputs. In this situation, the collaborative performance could drop well below human-expert level, with potentially disastrous consequences. This automation bias has been described with regard to poorly performing AI and other forms of automation⁴³.

Automation bias has been well-documented in non-medical areas, for example in the aviation industry. Investigations following air accidents have attributed pilot reliance on automated systems (in keeping with their training and company policy) as a primary cause of several crashes⁴⁴. A subtly different but related issue is that of 'situational awareness', and the effect thereon of automation⁴¹. As more and more aspects of information gathering are automated (for example, automatic measurement of

fetal head circumference), it is possible for the human operator to become disengaged from the task ('checking out'), leading to them to fail to interpret other pieces of information that are available to them (for example, that the brain has an abnormal appearance). A potential and concerning consequence of increased automation in ultrasound is that the sonographer role becomes more supervisory, with an expectation that the AI output will be largely accepted but constantly monitored. These mundane monitoring tasks are known to be very challenging and are often performed poorly by humans⁴⁵. This is a problem in non-medical areas, such as autonomous driving, and has not yet been solved definitively⁴⁶.

Finally, an overreliance on AI could lead to the deskilling of the sonographer workforce. Efficient use of specialist knowledge requires its frequent deployment, and lack of use makes the retrieval of such knowledge more difficult⁴⁵. Over time, use of AI systems could result in a deterioration of human performance on specific tasks. The repercussions could be exacerbated by the requirement for sonographers to take over when the AI model fails, which is likely to occur in the most challenging or complex scenarios⁴⁷. For example, asking a sonographer to measure biometry very rarely, but to do so only when the images are unclear, is unlikely to result in a high level of performance in these cases. Looking again to the aviation industry, the case of Air France Flight 447 in 2009 provides a sobering example of this effect: when asked to manually recover a disrupted flight path at high altitude after unexpected disconnection of the autopilot (a situation for which they had little practical or simulator experience), the pilots performed poorly, resulting in disaster⁴⁷.

Walking the tightrope: trust calibration

As shown in Table 1, the current state-of-the-art AI models developed for fetal ultrasound, although impressive in some contexts, do not work perfectly, and most have not been tested in prospective clinical trials. As with any system that is imperfect, there will be times that the AI models fail. This will produce an erroneous output that may be fed back to the sonographer, such as an incorrect biometric measurement, the saving of an incorrect standard plane or the misclassification of an image in terms of disease. This means that the sonographer will need to trust the AI system some of the time (otherwise there would be no point in using it), but also know when to doubt and over-rule it. In other words, for the collaborative approach between sonographers and AI shown in Figure 2 to be effective, sonographers must neither over-nor undertrust the AI system. This will require 'trust calibration', with a view to helping the sonographer interpret the AI output with an appropriate degree of trust or skepticism. Ideally, the sonographer should increase their trust in the AI system as the capability of the AI increases, and *vice versa* (Figure 3)^{29,48,49}.

Such trust calibration can be achieved by obtaining feedback on AI performance, and this has been explored in a variety of non-medical automation tasks^{29,48}. However,

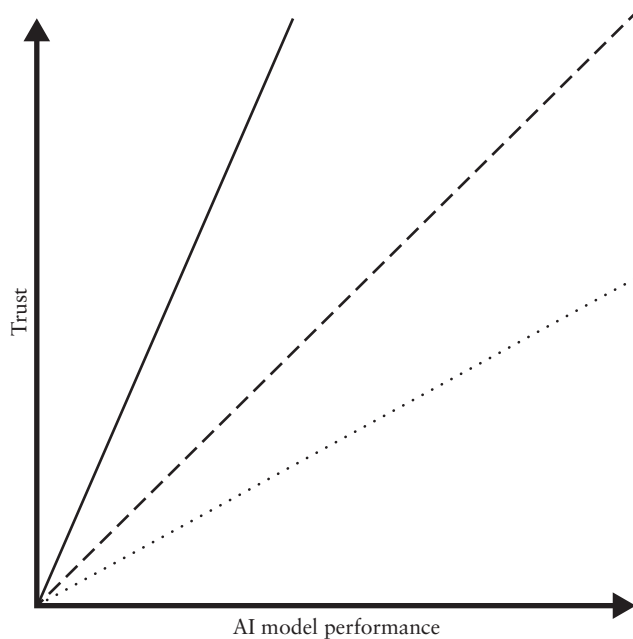


Figure 3 Relationship between sonographer trust in artificial intelligence (AI) model and model performance. —, overtrust: sonographer relies on poorly performing AI system; - - -, well-calibrated trust: appropriate level of trust for any given AI performance;, undertrust: sonographer disregards useful AI outputs. Adapted from Asan *et al.*⁴⁹.

application of these concepts to medical AI has received much less attention. Over time, a human user of a given AI system may learn the extent to which it produces reliable outputs. This is particularly challenging in the context of fetal ultrasound, as errors may not be detected until after the fetus is born. This long time interval would weaken the feedback loop to the sonographer, hindering good trust calibration. Nonetheless, in the absence of real-time feedback, assessing AI performance in this manner may be useful over a longer timescale to help human operators learn when and how to trust the AI system, and to highlight situations in which the AI system is likely to perform poorly.

Real-time estimation of model confidence may be helpful as an alternative approach, but this is generally challenging with machine-learning models. The confidence with which a deep neural network makes a prediction is not necessarily a reliable proxy for the likelihood of it being accurate. It is well-documented that neural networks may make an entirely incorrect prediction with a very high level of confidence, meaning that model confidence alone cannot be used for human trust calibration. There are potential complex mathematical solutions to this problem and this is an active area of research^{50,51}. However, as yet, no standard method to achieve this has been found in the machine-learning research community. If a robust metric of a given model's 'probability of correctness' could be derived in real time, it could guide the sonographer as to whether or not they should trust the AI output. This could be displayed in a variety of ways, including, for example, a traffic light system to give simple feedback. Taking this further, the AI output could be suppressed

and concealed from the sonographer if the likelihood of it being correct falls below a given threshold (known as 'selective prediction'⁵²). However, it is unlikely that any solution to this problem would work perfectly, and the problem of how to deal with incorrect AI outputs made with high confidence, and a high estimated probability of correctness, would remain in some circumstances.

Other forms of feedback might be useful in the calibration of sonographer trust. It may be possible to generate real-time graphical feedback of AI model output, with a form that varies according to the task performed (Figures 4 and 5). For a measurement task, an overlay depicting an automatically acquired biometric parameter might enable the sonographer to check that the correct anatomical area has been evaluated and that the measurement is accurate. Two examples of automated measurements, of which one seems appropriate and the other is clearly incorrect, are shown in Figures 4a and 4b, respectively. In the latter case, this visual output may indicate to the sonographer that the automatic measurement is untrustworthy and should be repeated manually. This method may be useful for measurements that are assessed readily in a visual manner, but may provide little benefit for more complex measurements, such as fetal myocardial strain, for which the accuracy of an AI output may be difficult to gauge, even with graphical feedback.

For classification tasks, such as disease detection, methods are available that generate a visual representation of the areas of the image that were most important in determining the AI output. This could help the sonographer to 'sense check' the output by assessing whether it was influenced predominantly by an area of the image that is clinically appropriate. One such approach is shown in Figure 5, in the form of gradient-weighted class activation maps (Grad-CAMs)⁵³ for a model that was trained to detect a specific form of congenital heart disease. We would expect the pixels of the image that correspond to the fetal heart to be most important to this model, and this is indeed the case in one example (Figure 5b). However, in the other example (Figure 5d), the main area of influence is outside of the fetus; this may be due to the model misinterpreting the image, since the relevant area looks similar to a heart in the model's feature composition space. This feedback might lead the sonographer to appropriately reject the output of this model. It is important to remember, however, that these outputs map only to the input features of an image, and do not represent the deep-layer dependencies of the AI model. They can be useful for checking that the right region of an image is being utilized for the overall model prediction, but not how the features depend upon one another, nor any other high-dimensional non-linear relationship.

Finally, in certain applications of AI, outputs may be generated simultaneously by an AI system and a human sonographer, allowing direct comparison. In many screening investigations, such as mammography, the use of two human readers is well-established. It has been suggested that AI could take the place of one of these readers, maintaining a high-quality screening

program while accommodating for the shortage of human experts⁵⁴. Double reading is uncommon in obstetric ultrasound at present, but could be facilitated by the introduction of AI. If AI models were running in the background, any discrepancy between the sonographer

output and that of the AI model could be flagged and investigated further.

Despite their limitations, all abovementioned types of feedback could help the sonographer to apply their clinical knowledge and experience when interpreting the output

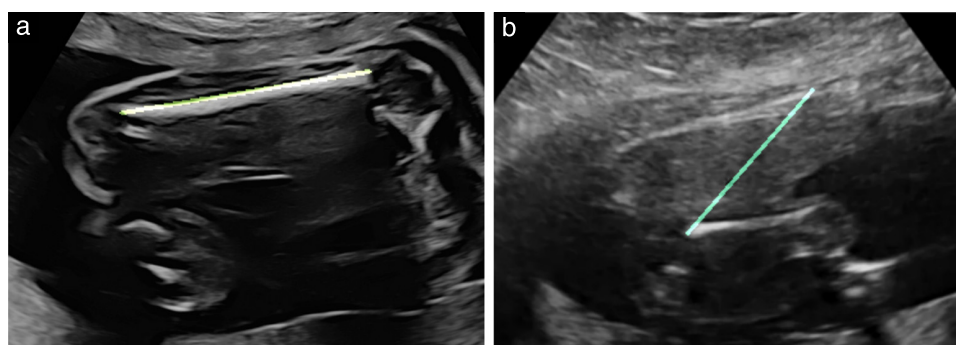


Figure 4 Grayscale ultrasound images annotated with real-time outputs (yellow and cyan lines) of artificial intelligence model trained to detect and measure femur length, as might be displayed to sonographer. (a) Appropriate start and end points for femur length are selected. (b) Inappropriate measurement points are selected, leading to inaccurate measurement of femur length.

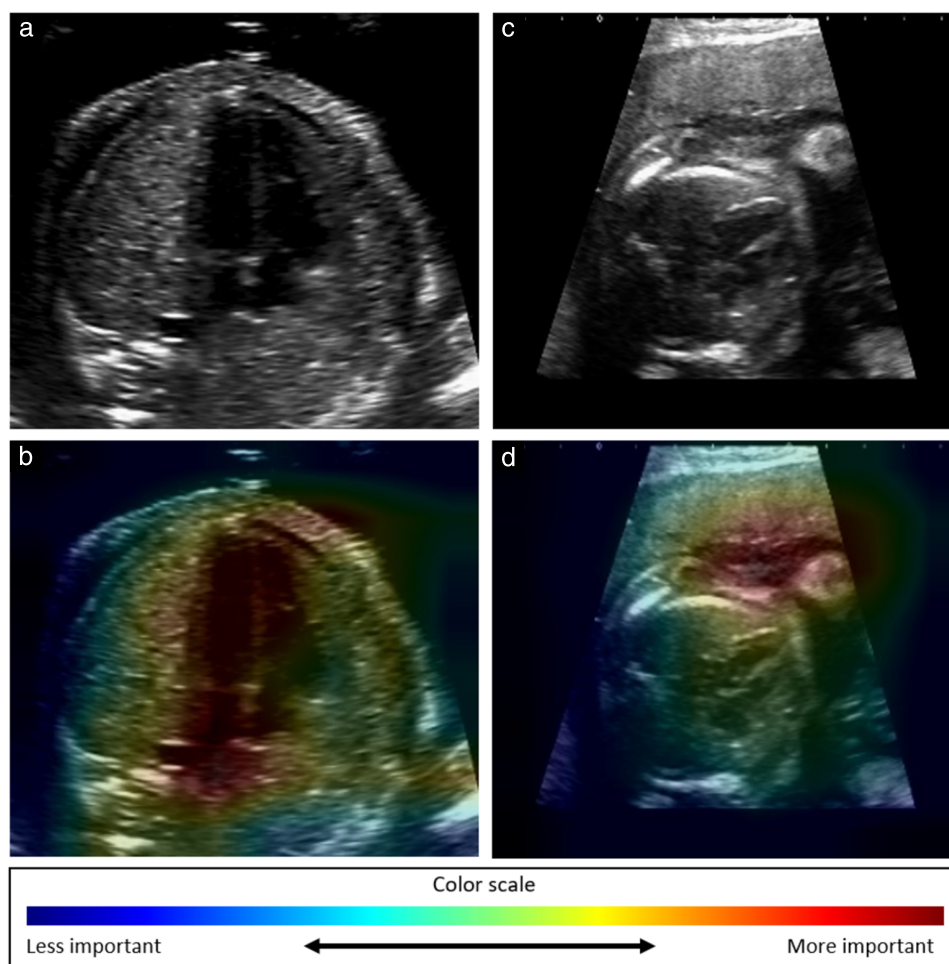


Figure 5 Unprocessed grayscale ultrasound images (a,c) and corresponding gradient-weighted class activation maps (Grad-CAMs) (b,d) of normal fetal heart in four-chamber view. Grad-CAMs are based on artificial intelligence model trained to classify hearts as normal *vs* those with a specific congenital heart lesion. Red and yellow colors indicate higher relative importance to final classification; green and blue colors indicate lower relative importance. (b) Grad-CAM of image in (a) shows appropriate focus on cardiac area. (d) Grad-CAM of image in (c) shows undesirable focus on area between placenta and thorax, which perhaps mimics appearance of fetal heart, meaning that model may be classifying inappropriately.

of an AI model. However, such feedback would have to be incorporated carefully into a collaborative system. Too much feedback could overwhelm, confuse or distract the sonographer, which may ultimately have a detrimental effect on the overall performance of the sonographer–AI team.

What would ideal collaboration look like?

Considering these various aspects, we can conceptualize the features that an ideal system of collaboration between the sonographer and a future AI system might possess. The AI system would need to work in real time and, of course, perform to a high level, with the sonographer able to trust and utilize its outputs. However, all AI systems have the potential to fail in at least some circumstances, and we would want the sonographer to be able to identify when such an event had happened and to disregard the model output. This trust calibration could be achieved in myriad ways, depending on the nature of the process being automated, but it would be essential for any relevant information to be fed back to the sonographer in a way that could be assessed rapidly and confidently, without distracting from the scan itself. Empirical work will be required to develop and test these user interfaces, and to determine how AI can be integrated and feedback delivered in the most effective and efficient way.

Conclusion

The application of AI to fetal ultrasound imaging has already entered clinical practice, and its role is likely to increase rapidly over the coming years. The real-time nature of ultrasound evaluation means that fully autonomous scanning is an unlikely prospect and, therefore, sonographers and AI systems will need to collaborate for the benefit of patients. Harnessed appropriately, AI has the potential to improve many aspects of fetal ultrasound, but researchers and clinicians need to be aware of the dangers of both over-reliance on and aversion to AI. Accurate calibration of trust in AI systems is likely to prove integral to the successful translation of AI tools, and the development of usable metrics and feedback methods to facilitate this will be crucial.

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